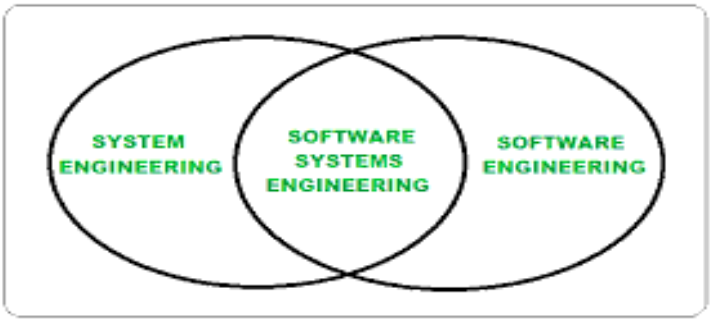
Unit VI

System Engineering

Systems Engineering is an engineering field that takes an interdisciplinary approach to product development. Systems engineers analyze the collection of pieces to make sure when working together, they achieve the intended objectives or purpose of the product. For example, in automotive development.



COMPUTER-BASED SYSTEMS

Definition: a computer-based system as A set or arrangement of elements that are organized to accomplish some predefined goal by processing information.

The goal may be to support some business function or to develop a product that can be sold to generate business revenue. To accomplish the goal, a computer-based system makes use of a variety of system elements.

Software: Computer programs, data structures, and related documentation that serve to affect the logical method, procedure, or control that is required.

Hardware : Electronic devices that provide computing capability, the inter[1]connectivity devices (e.g., network switches, telecommunications

devices) that enable the flow of data, and electromechanical devices (e.g., sensors, motors, pumps) that provide external world function.

People: Users and operators of hardware and software.

Database: A large, organized collection of information that is accessed via software.

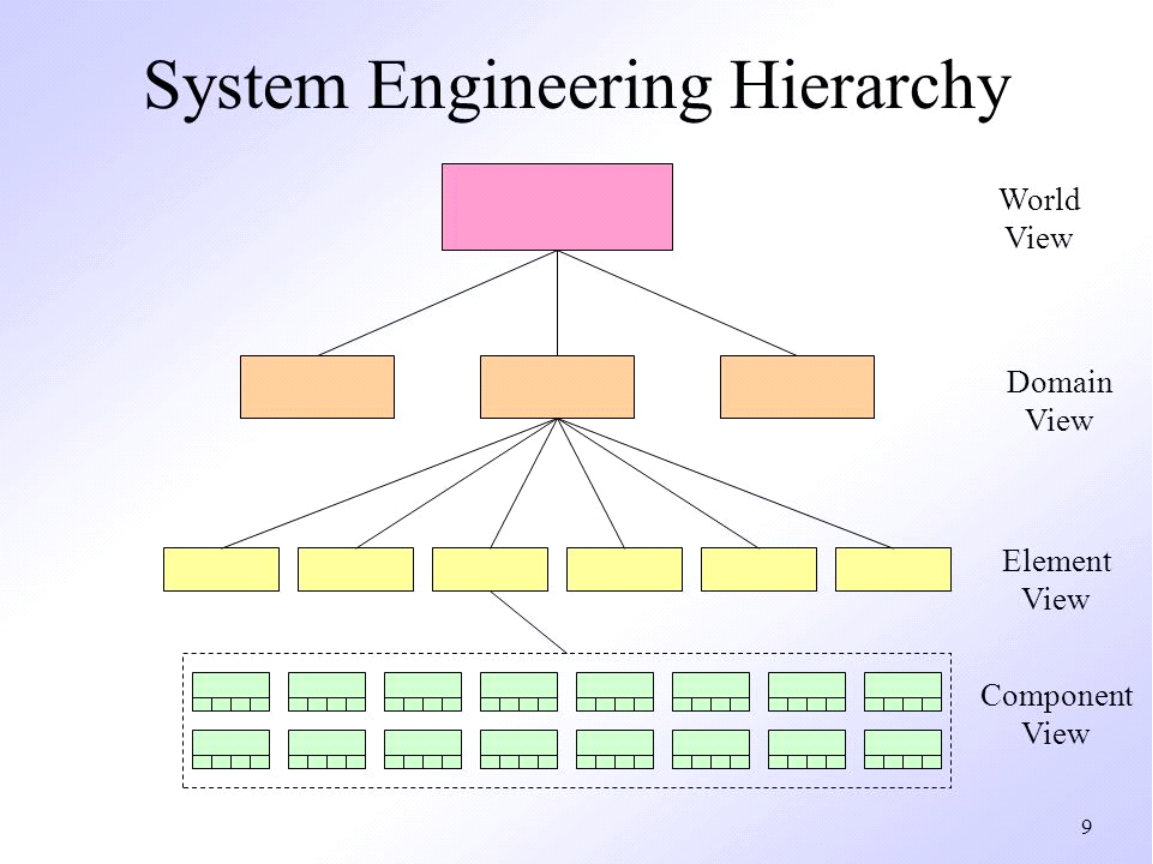
Documentation: Descriptive information (e.g., hardcopy manuals, on-line help files, Web sites) that portrays the use and/or operation of the system.

Procedures: The steps that define the specific use of each system element or the procedural context in which the system resides.

To summarize, the manufacturing cell and its macro elements each are composed of system elements with the generic labels: software, hardware, people, database, procedures, and documentation.

THE SYSTEM ENGINEERING HIERARCHY

The role of the system engineer is to define the elements for a specific computer[1]based system in the context of the overall hierarchy of systems (macro elements).



The system engineering process usually begins with a “world view.” That is, the entire business or product domain is examined to ensure that the proper business or technology context can be established. The world view is refined to focus more fully on specific domain of interest. Within a specific domain, the need for targeted system elements (e.g., data, software, hardware, people) is analyzed. Finally, the analysis, design, and construction of a targeted system element is initiated. At the top of the hierarchy, a very broad context is established and, at the bottom, detailed technical activities, performed by the relevant engineering discipline (e.g., hardware or software engineering), are conducted.

SYSTEM MODELING

System engineering is a modeling process. Whether the focus is on the world view or the detailed view, the engineer creates models that -

• Define the processes that serve the needs of the view under consideration.

• Represent the behavior of the processes and the assumptions on which the behavior is based.

• Explicitly define both exogenous and endogenous input3 to the model.

• Represent all linkages (including output) that will enable the engineer to better understand the view

To construct a system model, the engineer should consider a number of restraining factors:

1**.Assumptions** that reduce the number of possible permutations and variations, thus enabling a model to reflect the problem in a reasonable manner.

2.**Simplifications** that enable the model to be created in a timely manner. To illustrate, consider an office products company that sells and services a broad range of copiers, faxes, and related equipment.

3**.Limitations** that help to bound the system.

4**.Constraints** that will guide the manner in which the model is created and the approach taken when the model is implemented.

5.**Preferences** that indicate the preferred architecture for all data, functions, and technology.

SYSTEM SIMULATIONS

Many computer-based systems interact with the real world in a reactive fashion. That is, real-world events are monitored by the hardware and

software that form the computer-based system, and based on these events, the system imposes control on the machines, processes, and even people who cause the events to occur. Real-time and embedded systems often fall into the reactive systems category.

Unfortunately, the developers of reactive systems sometimes struggle to make them perform properly. Until recently, it has been difficult to predict the performance, efficiency, and behavior of such systems prior to building them.

In a very real sense, the construction of many real-time systems was an adventure in "flying." Surprises (most of them unpleasant) were not discovered until the system was built and "pushed off a cliff." If the system crashed due to incorrect function, inappropriate behavior, or poor performance, we picked up the pieces and started over again.

Many systems in the reactive category control machines and/or processes (e.g., commercial aircraft or petroleum refineries) that must operate with an extremely high degree of reliability. If the system fails, significant economic or human loss could occur. For this reason, the approach described by Graham is both painful and dangerous

Today, software tools for system modeling and simulation are being used to help to eliminate surprises when reactive, computer-based systems are built. These tools are applied during the system engineering process, while the role of hardware and software, databases and people is being specified. Modeling and simulation tools enable a system engineer to "test drive" a specification of the system.

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